

Soumyajit Karmakar

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 [soumyajitkarmakar.github.io](https://github.com/soumyajitkarmakar) •  Scholar

EDUCATION

University of California San Diego

2024 - 2026

Master of Science in Computer Science.

Thesis Advisor: [Prof. Mikhail Belkin](#).

GPA: 3.9/4.0

Indian Institute of Information Technology, Guwahati

2019 - 2023

Bachelor of Technology in Computer Science and Engineering.

CGPA: 9.35/10.00 (Silver medalist)

RESEARCH EXPERIENCE

University of California San Diego

Graduate Research Assistant

Advisor: [Prof. Mikhail Belkin](#)

September 2024 - Present

- Using Recursive Feature Machines to understand the internal activation geometry of Large Language Models (LLMs).
- Analyzing the structure of learned representations to find universal linear maps that can transform LLM steering vectors across antonyms, layers, and languages, allowing algebraic synthesis of new steering vectors without re-running extraction.

Fujitsu Research & Indian Institute of Science, Bengaluru

Advisor : [Prof. R. Venkatesh Babu](#)

September 2023 - August 2024

- Designed a prompt-based inference mechanism using CLIP for robust person re-identification, eliminating the need for extensive fine-tuning.

University of North Carolina at Charlotte, North Carolina

Research Intern - Advisor : [Dr. Srijan Das](#) and [Dr. Michael S. Ryoo](#)

August 2022 - March 2023

- Worked on enhancing Vision Transformer (ViT) generalization on small datasets by integrating Masked Autoencoders (MAE) as a self-supervised auxiliary task.
- Work led to paper at WACV 2024.

ACHIEVEMENTS

- **Global Rank 1**, in the Parameterized Algorithms and Computational Experiments (PACE) 2022 (Heuristic Track). [Link](#) for the paper. [Link](#) for the solver.

PUBLICATIONS

- **Soumyajit Karmakar**, Neil M., Mikhail B. “Relational Structure Between Steering Vectors in Large Language Models”. (under review)
- **Soumyajit Karmakar***, Shrinivas R.*, Harsh R.*, Yasuhiro A., Genta S., Venkatesh B. “Prompt-ReID: Prompt Based Inference for Robust Re-Identification”. (*Arxiv*)
- Srijan D., Tanmay J., Dominick R., Pranav B., **Soumyajit Karmakar**, Shyam M., Xiang L., Abhijit D., Michael R. “Limited Data, Unlimited Potential: A Study on ViTs Augmented by Masked Autoencoders”. In Proceedings of the 2024 **WACV**, Waikoloa, Hawaii, USA.
- **Soumyajit Karmakar**, Abeer B., Prashant G., Sumeet S., Sanjay S. “Convolutional Ensembling based Few-Shot Defect Detection Technique”. In Proceedings of the 2022 **ICVGIP**, IIT Gandhinagar, India.
- Aman J., Sachin A., Nimish A., **Soumyajit Karmakar**, Srinibas S. “Feedback vertex set using Edge Density and REmove Redundant (FEDRER): A heuristic solver for finding a feedback vertex set in a directed graph” . In poster session of the 2022 **IPEC**, Potsdam, Germany.

Technical Skills: Python, PyTorch, JAX, C++, LaTeX, Git.